Wavelet-Based Signal Extraction and Regime Detection in Commodity Futures: A Rigorous Walk-Forward Study

Abstract

This report develops, tests, and critically evaluates a wavelet-based trading framework for commodity futures. Starting from the continuous wavelet transform (CWT) as introduced by Torrence and Compo [1], we build a fully causal, walk-forward pipeline for extracting band-limited cyclical components from commodity log-returns, gate their use by formal statistical significance against a red-noise null, and combine them with a Hidden Markov Model (HMM) regime classifier and cross-wavelet analysis against macroeconomic drivers. Every stage is subjected to an out-of-sample discipline: bootstrap confidence intervals, train/test band selection, and rolling multi-window validation. The central empirical finding is not that the wavelet signal generates a stable, exploitable alpha on its own – it does not, once look-ahead bias and multiple-testing effects are controlled for – but that wavelet-detected cyclical power is strongly and independently confirmed (via HMM) to concentrate in high-volatility market regimes, and that this same information, used as an *inverse* risk-reduction overlay rather than a directional bet, materially improves the risk-adjusted profile of a diversified mean-reversion ensemble while *failing* on a directional buy-and-hold exposure – a distinction with a clear economic explanation. The report is organized to first establish the theory, then the methodology, then the validation framework, and finally the empirical results and their interpretation.

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1 Introduction

Commodity prices are not simply random walks corrupted by noise. Superimposed on long-run supply-and-demand trends and high-frequency microstructure noise, there is often a genuine cyclical component: storage economics create a restoring force in energy prices, seasonal harvest and demand patterns create annual cycles in agricultural commodities, and positioning/flow dynamics create shorter oscillations of a few weeks to a few months. The central empirical question of this report is simple to state and hard to answer honestly: *can a signal-processing tool that isolates these cycles be turned into a trading edge, and if so, under what conditions?*

The classical tool for decomposing a signal into frequency components is the Fourier transform, but it has a well-known limitation for financial time series: it assumes stationarity, returning a single frequency content averaged over the *entire* sample. A cycle that was strong in 2016 and absent in 2022 is smeared into a single, largely uninformative average. The windowed Fourier transform (WFT) attempts to fix this by sliding a fixed-length window across the series, but this imposes an arbitrary resolution trade-off that does not adapt to the frequency being analyzed [1].

Wavelet analysis solves this by using a basis function that is simultaneously localized in time *and* frequency, with a resolution that automatically adapts to scale: good time resolution at short cycles, good frequency resolution at long cycles. This report uses wavelet analysis in exactly the sense developed by Torrence and Compo [1] for geophysical time series (their worked example is the El Niño–Southern Oscillation), transplanted to commodity futures, with all the statistical machinery (significance testing, cross-wavelet analysis, coherence) carried over essentially unchanged.

The report is structured as follows. Section 2 develops the theory of the continuous wavelet transform, its statistical significance testing, and its cross-wavelet extensions, in reasonable technical depth but without requiring prior exposure to time-frequency analysis. Section 3 discusses why and how wavelets have been used in finance and what is genuinely different about applying them

to *trading* rather than *description*. Section 4 describes the full methodology built for this study, including the causal walk-forward architecture that is the technical core of the exercise. Section 5 describes the validation framework – bootstrap, out-of-sample band selection, rolling validation, ensembling – built specifically to guard against the well-documented tendency of backtested strategies to look good by construction. Section 6 covers two extensions: a cross-wavelet analysis against macroeconomic drivers, and a regime-detection layer based on a Hidden Markov Model. Section 7 reports the empirical results on WTI crude oil futures (CL=F). Section 8 discusses what the results actually mean, and Section 9 concludes.

2 Theoretical Background: Wavelet Analysis

2.1 From Fourier to Time-Frequency Analysis

Given a discrete time series x_n, $n = 0, \ldots, N-1$, sampled at interval δt, the discrete Fourier transform is

$$\hat{x}_k = \frac{1}{N} \sum_{n=0}^{N-1} x_n \, e^{-2\pi i k n / N}, \qquad k = 0, \ldots, N-1. \tag{1}$$

The resulting power spectrum $|\hat{x}_k|^2$ describes how much variance is attributable to each frequency k, averaged over the *whole* sample. If the process generating x_n is non-stationary – as almost all financial and commodity return series genuinely are – this average can be misleading: a strong short-lived cycle and a weak long-lived one can produce similar Fourier peaks.
The windowed Fourier transform addresses this by computing the transform on a sliding segment of fixed length T. This reintroduces localization in time, but at the cost of an *arbitrary and fixed* time-frequency resolution: a window short enough to resolve fast oscillations is too short to resolve slow ones, and vice versa [1].

2.2 The Continuous Wavelet Transform

The continuous wavelet transform (CWT) resolves this trade-off by using a localized oscillatory function $\psi_0(\eta)$ – the *mother wavelet* – that is dilated and translated:

$$W_n(s) = \sum_{n'=0}^{N-1} x_{n'} \, \psi_0^* \left(\frac{(n' - n)\, \delta t}{s} \right), \tag{2}$$

where s is the wavelet *scale* (playing the role of an inverse frequency), n is the localized time index, and $*$ denotes complex conjugation. For ψ_0 to be an admissible wavelet it must have zero mean and be well localized in both time and frequency [1]. Varying s while sliding n across the series produces a full time-scale decomposition: unlike the WFT, the resolution now *automatically adapts* to s – narrow (good time resolution) at small scales, wide (good frequency resolution) at large scales. In practice the sum above is never evaluated directly; it is computed far more efficiently as an inverse FFT of the product of the Fourier transforms of x_n and ψ_0, exploiting the convolution theorem.

2.3 The Morlet Wavelet

Among the admissible choices, this study uses the **Morlet wavelet**, a plane wave modulated by a Gaussian envelope:

$$\psi_0(\eta) = \pi^{-1/4} e^{i\omega_0 \eta} e^{-\eta^2/2}, \tag{3}$$

where ω_0 is a non-dimensional frequency, conventionally set to 6 to satisfy the admissibility condition while giving a good balance between time and frequency localization [1]. Two properties make the Morlet wavelet the natural choice for a financial application:

- It is **complex-valued**, so the wavelet transform $W_n(s)$ carries both an amplitude $|W_n(s)|$ and a phase $\tan^{-1}[\Im\{W_n(s)\}/\Re\{W_n(s)\}]$. The amplitude measures how strong a given cycle is; the phase, used later in the cross-wavelet analysis, measures the lead-lag relationship between two series.
- Its scale s is (for $\omega_0 = 6$) almost exactly equal to the equivalent Fourier period, $\lambda \approx 1.03\, s$, which makes the scale axis directly interpretable in calendar time (e.g. "a 30-day cycle") rather than requiring a separate conversion table [1].

Real-valued alternatives (e.g. the Derivative-of-Gaussian family) trade away the phase information in exchange for sharper localization of discontinuities; since this study is explicitly interested in lead-lag relationships (Section 2.8), the complex Morlet is preferred throughout.

2.4 Wavelet Power Spectrum and Scale Selection

The **wavelet power spectrum** is $|W_n(s)|^2$. Scales are chosen as fractional powers of two,

$$s_j = s_0\, 2^{j\delta j}, \qquad j = 0, 1, \ldots, J, \tag{4}$$

where s_0 is the smallest resolvable scale (set to twice the sampling interval, $s_0 = 2\,\delta t$) and J determines the number of octaves spanned. In this study $\delta j = 1/12$ (twelve sub-octaves per octave) and J spans seven octaves, giving fine and roughly log-uniform coverage of periods from a few days to several years.

2.5 The Cone of Influence

Because the wavelet transform in practice operates on a finite, zero-padded series, the transform near the beginning and end of the sample is contaminated by the artificial padding, and increasingly so at larger scales (since the wavelet footprint widens with s). The **cone of influence** (COI) is the region where this edge distortion becomes important, formally defined as the e-folding time of the wavelet's autocorrelation at each scale [1]. Any interpretation of wavelet power inside the COI must be treated with caution.
This is not a cosmetic detail: it becomes the central engineering constraint of the causal signal-extraction procedure in Section 4.3, since *by construction* every real-time estimate sits exactly at the right edge of its own COI.

2.6 Statistical Significance Against Red Noise

A raw wavelet power spectrum is, on its own, uninformative: financial return series have non-trivial autocorrelation even under the null of "no genuine cycle", and any transform of noisy data produces peaks and troughs that can be mistaken for structure. Torrence and Compo [1] address this by deriving the theoretical wavelet power spectrum expected under a **red noise** null – a lag-1 autoregressive process,

$$x_n = \alpha\, x_{n-1} + z_n, \qquad z_n \sim \mathcal{N}(0, 1), \tag{5}$$

whose theoretical Fourier power spectrum is

$$P_k = \frac{1 - \alpha^2}{1 + \alpha^2 - 2\alpha \cos(2\pi k/N)}. \tag{6}$$

Under the assumption that the wavelet coefficients are (locally) normally distributed, the local wavelet power spectrum is distributed as

$$\frac{|W_n(s)|^2}{\sigma^2} \;\Rightarrow\; \tfrac{1}{2}\, P_k\, \chi^2_2, \tag{7}$$

where σ^2 is the series variance and χ^2_2 is the chi-square distribution with two degrees of freedom (one for real-valued wavelets). Choosing a confidence level (95% in the classical formulation) and inverting the χ^2_2 distribution gives a scale-dependent threshold against which the observed power can be tested. This is a **closed-form** test – no Monte Carlo simulation is required – which turns out to matter a great deal for the computational feasibility of the walk-forward procedure developed later.

A useful refinement, since a single point in the time-scale plane is a noisy estimate, is to **average over a band of scales** $[s_{j_1}, s_{j_2}]$ (Torrence & Compo's equation 24), giving the scale-averaged power

$$\overline{W}^2_n = \frac{\delta j\, \delta t}{C_\delta} \sum_{j=j_1}^{j_2} \frac{|W_n(s_j)|^2}{s_j}, \tag{8}$$

where C_δ is a wavelet-dependent reconstruction constant ($C_\delta = 0.776$ for the Morlet wavelet with $\omega_0 = 6$). This scale-averaged power is itself χ^2_ν-distributed, with an effective number of degrees of freedom ν that depends on how many (correlated) scales are averaged [1]. This is the exact test used throughout Section 4 to decide, at each point in time, whether the cyclical power in the traded band is distinguishable from red noise.

2.7 Reconstruction and Band-Pass Filtering

Since the CWT is a redundant, overcomplete representation, the original series can be reconstructed from a *subset* of scales, giving a band-pass filtered version of the series that retains only the oscillatory content in a chosen period range:

$$x'_n = \frac{\delta j\, \delta t^{1/2}}{C_\delta\, \psi_0(0)} \sum_{j=j_1}^{j_2} \frac{\Re\{W_n(s_j)\}}{s_j^{1/2}}. \tag{9}$$

This is the mechanism used throughout this report to isolate "the 20-to-60-day cycle" (or any other band) from a commodity return series: it is nothing more than a sum over a restricted range of scales in the inverse transform.

2.8 Cross-Wavelet Transform and Wavelet Coherence

Given two series X and Y with wavelet transforms $W^X_n(s)$ and $W^Y_n(s)$, the **cross-wavelet transform** is

$$W^{XY}_n(s) = W^X_n(s)\, W^{Y*}_n(s), \tag{10}$$

a complex quantity whose modulus $|W^{XY}_n(s)|$ measures the common power between the two series at scale s and time n, and whose phase measures their lead-lag relationship in that band. Torrence and Compo [1] derive a **closed-form** significance test for the cross-wavelet power, analogous to Eq. (7) but based on the product of two chi-square-distributed variables, whose cumulative distribution involves the modified Bessel function K_0 (their equation 30). Crucially, this significance test requires *no Monte Carlo simulation*: it is evaluated directly from the two series' individual AR(1) parameters.

Wavelet coherence, by contrast, is defined as

$$R^2_n(s) = \frac{\left|\left\langle s^{-1} W^{XY}_n(s) \right\rangle\right|^2}{\left\langle s^{-1}|W^X_n(s)|^2 \right\rangle \left\langle s^{-1}|W^Y_n(s)|^2 \right\rangle}, \tag{11}$$

where $\langle\cdot\rangle$ denotes smoothing in time and scale. Without this smoothing, coherence is identically one everywhere by construction – a point noted by Liu (1994) and discussed by Torrence and Compo [1]. Unlike the cross-wavelet power test, wavelet coherence has **no known closed-form null distribution**: significance must be assessed via Monte Carlo simulation of surrogate AR(1) pairs [2]. This asymmetry – cross-wavelet power is cheap (closed-form), coherence is expensive

(Monte Carlo) – is not a minor implementation detail: it directly determines which of the two tools is computationally viable inside a walk-forward loop re-estimated hundreds of times, as discussed in Section 6.1.

3 Wavelets in Finance and Trading

3.1 Why Finance Is a Natural Domain for Wavelets

Financial and commodity time series share three properties that make them good (if challenging) candidates for wavelet analysis:

1. **Non-stationarity.** Volatility, trend, and cyclical structure all evolve over time – regime shifts, structural breaks, and policy changes are the norm, not the exception.

2. **Multi-horizon structure.** Different market participants operate at genuinely different frequencies (intraday market makers, swing traders, macro funds, strategic reserve managers), so a market series can legitimately contain superimposed cycles at very different scales simultaneously.

3. **Fat tails and jumps.** Wavelets, being localized in time, are naturally suited to characterizing transient, localized events (a jump or a volatility burst) without smearing their signature across the whole sample the way a Fourier transform would.

This has motivated a substantial literature applying wavelet methods to economics and finance, including textbook treatments by Gençay, Selçuk, and Whitcher [3] and In and Kim [4], and a broader methodological survey by Ramsey [5]. Typical applications include decomposing volatility by horizon, studying the scale-dependent relationship between markets (e.g. via wavelet coherence, exactly as in Section 2.8), and testing whether return predictability is horizon-specific rather than uniform.

3.2 From Description to Prediction: The Causality Problem

There is a critical and easily overlooked gap between using wavelets *descriptively* – to characterize the historical spectral content of a series, exactly as Torrence and Compo [1] do for the Niño3 SST record – and using them to build a **trading signal**. The former is legitimately computed once over the full available history: it is a retrospective statement about what happened. The latter must never use information that would not have been available at the time the signal is generated.

The Morlet wavelet's footprint is symmetric in time: the coefficient $W_n(s)$ at time n is influenced by data both *before and after* n inside the transform's support. A full-sample CWT reconstruction (Eq. (9), evaluated once over the whole series) therefore contains a subtle but real form of **look-ahead bias** at every interior point: the reconstructed value at time n "knows" about returns that occurred after n. This is invisible in a purely descriptive plot, but fatal for anything claiming to be a real-time trading rule.

The methodology in Section 4 is built specifically to close this gap, at the cost of being considerably more expensive to compute than a single full-sample transform.

4 Methodology

4.1 Data and Preprocessing

The base asset used throughout is WTI crude oil futures (Yahoo Finance ticker `CL=F`), sampled daily. Two preprocessing decisions deserve explicit justification:

Working with log-returns, not price levels. Commodity price levels are far too non-stationary for the linear detrending typically sufficient in geophysical applications: WTI crude experienced a

multi-year price collapse (2014–2016), the COVID-19 demand shock and the unprecedented *negative* settlement price of 20 April 2020, and the 2022 supply shock following the invasion of Ukraine. A single linear detrend does not come close to stationarizing a series with this much structural change, and attempting to fit an AR(1) red-noise background (Eq. (6)) on such a series is numerically unstable. Working instead with daily log-returns,

$$r_n = \ln p_{n+1} - \ln p_n, \tag{12}$$

removes the need for detrending almost entirely and produces a substantially more well-behaved series for both the CWT and its red-noise null.

Cleaning non-positive price ticks. The negative WTI settlement of 20 April 2020 is a well-documented artifact of front-month futures delivery mechanics, not an economically meaningful price observation for return-based analysis; log-returns are undefined across a sign change. Non-positive ticks are forward-filled from the last valid observation before computing returns.

4.2 Robust Estimation of the Red-Noise Background

The lag-1 autocorrelation α needed for the red-noise null (Eq. (6)) is normally estimated via a bias-corrected (Kendall-type) estimator. This estimator's closed-form solution involves a quadratic equation whose discriminant can become negative on short or strongly non-stationary windows, causing the estimation to fail outright. A robust wrapper is used throughout: the bias-corrected estimator is attempted first, and on failure the estimation falls back to the plain sample lag-1 autocorrelation, clipped to $[0, 0.99]$ for numerical safety.

4.3 Causal Walk-Forward Signal Extraction

To eliminate the look-ahead bias discussed in Section 3.2, the band-pass reconstruction of Eq. (9) is **never** computed once over the full sample for trading purposes. Instead, the CWT is periodically *re-estimated* using only the data available up to each point in time, and only the **endpoint** of each such reconstruction – the most recent value – is retained as the live signal. Algorithm 1 formalizes this procedure.

Algorithm 1 Causal walk-forward band-pass signal

Require: log-return series $r_1, \ldots, r_N$; band $[T_1, T_2]$; warm-up length L; refit interval Δ

signal $\leftarrow \mathbf{0}_N$, strength $\leftarrow \mathbf{0}_N$
for $i = L, L + \Delta, L + 2\Delta, \ldots, N$ **do**
  $w \leftarrow (r_1, \ldots, r_i)$ ▷ expanding window, no future data
  $W, s, \text{period} \leftarrow \text{CWT}(w)$ ▷ Morlet CWT of the window only
  $\alpha \leftarrow \text{RobustAR1}(w)$
  $x' \leftarrow$ reconstruction of w restricted to scales with period $\in [T_1, T_2]$ ▷ Eq. (9)
  $\rho \leftarrow$ scale-averaged power ratio at the window's last point vs. its significance threshold ▷ Eqs. (8)–(7)
  for $t = i, \ldots, \min(i + \Delta, N)$ **do**
    signal$[t] \leftarrow x'_i$ ▷ hold the endpoint value until the next refit
    strength$[t] \leftarrow \text{clip}(\rho, 0, 1)$
  end for
end for
return signal, strength

Because the window used at step i never extends past i, the resulting signal is provably causal: no observation dated after i ever enters the computation of signal$[i]$. The unavoidable cost is that the reconstructed value is always evaluated exactly at the edge of its own cone of influence (Section 2.5), and is therefore intrinsically noisier than the equivalent point in a full-sample, non-causal analysis – a real and permanent limitation of real-time wavelet estimation, not an implementation defect.

4.4 From Signal to Position: Mean-Reversion Rule

The band-passed oscillator x'_n from Algorithm 1 is converted into a trading signal via a rolling z-score,

$$z_n = \frac{x'_n - \mu_n}{\sigma_n}, \tag{13}$$

with μ_n, σ_n computed on a trailing window, and a **hysteresis** entry/exit rule to avoid excessive turnover from noise around a single threshold:

$$\text{state}_n = \begin{cases} -1 & \text{if state}_{n-1} = 0 \text{ and } z_n > z_{\text{entry}} \\ +1 & \text{if state}_{n-1} = 0 \text{ and } z_n < -z_{\text{entry}} \\ 0 & \text{if } |\text{state}_{n-1}\, z_n| < z_{\text{exit}} \\ \text{state}_{n-1} & \text{otherwise.} \end{cases} \tag{14}$$

This is a conventional discretionary mean-reversion rule; the methodological content of this report lies not in the rule itself but in how the underlying oscillator is constructed and gated.

4.5 Continuous Significance-Based Position Scaling

An initial version of this study used a **binary** gate: new positions were only allowed when the scale-averaged power ratio ρ (step 6 of Algorithm 1) exceeded 1, i.e. when the band was significant at the chosen confidence level. Empirically, this gate fired only 3–4% of the time even at a relaxed 80% confidence level, leaving too few trades in any reasonably sized out-of-sample window to draw a meaningful conclusion (Section 5). Genuine, statistically detectable cyclicality in a narrow frequency band of daily commodity returns is, empirically, rare – consistent with a near-efficient market, where any *persistent* exploitable periodicity would already have been arbitraged away.
The gate was therefore replaced by a **continuous** multiplier, $\text{clip}(\rho, 0, 1)$, applied directly to position size rather than as an entry/exit switch. This preserves the same statistical logic – weak or absent cyclicality receives less size – while avoiding the all-or-nothing collapse of the binary version.

4.6 Risk Management: Volatility Targeting

Independently of the wavelet-based sizing above, position size is scaled to target a constant annualized volatility $\sigma^\star$ (20% in this study), using a trailing realized-volatility estimate computed on **past** returns only (shifted by one day, so the sizing decision itself introduces no look-ahead):

$$\text{size}_n = \min\left(\frac{\sigma^\star}{\widehat{\sigma}_{n-1}},\, L_{\max}\right), \tag{15}$$

capped at a maximum leverage $L_{\max} = 2$. This is standard risk-engineering practice, orthogonal to the wavelet methodology, but essential in practice: an early version of the backtest without this control suffered a near-total drawdown driven by a single extreme return around the 20 April 2020 event, illustrating how a single data anomaly can dominate an otherwise reasonable signal if position sizing is left unmanaged.

4.7 Backtest Engine

Given a position series $\text{pos}_n \in [-L_{\max}, L_{\max}]$, the strategy's net daily return is

$$\tilde{r}_n = \text{pos}_{n-1}\, r_n \;-\; c\,|\text{pos}_n - \text{pos}_{n-1}|, \tag{16}$$

where c is a proportional transaction cost (2 basis points round-trip in this study) and the one-period lag on pos ensures the position is only exposed to returns realized *after* it was established. All performance statistics reported below (annualized return, volatility, Sharpe ratio, maximum drawdown, Calmar ratio) are computed from $\tilde{r}_n$.

5 Validation Framework

A backtest constructed with the discipline of Section 4 still says nothing, on its own, about whether an observed positive Sharpe ratio reflects genuine structure or overfitting to a particular sample. Four complementary tools are used to address this.

5.1 Block Bootstrap Confidence Intervals

Financial returns are autocorrelated (through volatility clustering and, here, through the deliberately constructed cyclical signal itself), so an i.i.d. bootstrap would understate the true sampling variability of the Sharpe ratio. A **circular block bootstrap** is used instead: overlapping blocks of $b = 20$ consecutive daily returns are resampled with replacement (wrapping around the end of the sample) to build up synthetic return paths of the original length, from which an empirical distribution of the annualized Sharpe ratio is obtained. The 5th and 95th percentiles of this distribution give a 90% confidence interval, and the fraction of bootstrap draws with positive Sharpe gives an approximate one-sided significance measure.

5.2 Out-of-Sample Band Selection

Section 4 leaves one free parameter unresolved: which period band $[T_1, T_2]$ should be traded. Selecting this band by maximizing its Sharpe ratio over the *entire* available sample and then reporting that same Sharpe ratio is a textbook case of **data snooping**. The correct procedure – used throughout this study whenever a band or driver must be chosen – is:

1. Fix a split date; compute each candidate band's Sharpe ratio using **only** data before the split ("train").
2. Select the band with the highest train-period Sharpe ratio.
3. Report that band's performance **only** on the data after the split ("test"), which never influenced the selection.

Any discrepancy between train and test performance for the selected band is direct, honest evidence about how reliable the selection procedure is – as opposed to how good the best-*possible* band happened to be in hindsight.

5.3 Rolling Multi-Window Validation

A single train/test split can be a lucky or unlucky artifact of the chosen date. The procedure above is therefore repeated across nine rolling annual windows (June of each year from 2016 to 2024, each followed by a 12-month test period), tracking, for each window: (i) the Sharpe ratio realized by the band chosen on that window's train data, and (ii) how that realized Sharpe ranks against what *every* candidate band in the grid actually achieved in the same test window (its rank would be 1 if the selection procedure had picked the true best performer). Across many independent windows, a procedure with genuine predictive power should rank in the top half of the grid *more* often than chance, and should have a positive test Sharpe *more* often than a coin flip.

5.4 Ensemble Diversification Across Bands

Where the rolling validation above reveals that *some* band in the grid performs well in almost every window, but that the ex-ante selection procedure cannot reliably identify *which one* in advance, the natural mitigation is not to keep searching for a better selection rule, but to **diversify away the selection risk entirely**: an equal-weight combination of all candidate bands' daily returns is held simultaneously, requiring no ex-ante choice at all. This ensemble is compared, window by window, against the single ex-ante pick, both in Sharpe ratio and in how consistently it reduces drawdown relative to the pick.

6 Extensions

6.1 Cross-Wavelet Analysis Against Macroeconomic Drivers

Section 2.8 noted that the cross-wavelet power test is closed-form, unlike wavelet coherence, which requires an expensive Monte Carlo simulation (of order minutes per pair, in this implementation, for a single full-sample estimate). This asymmetry is decisive for a walk-forward application: re-estimating a Monte Carlo coherence test at every one of several hundred refit points is computationally impractical, whereas the closed-form cross-wavelet significance test is cheap enough to call at every refit.

This study therefore builds a causal, walk-forward analogue of Algorithm 1 using the **cross-wavelet** amplitude between the commodity and a candidate macroeconomic driver (the USD index, the US 10-year yield, the VIX, and the S&P 500 index, each tested independently), in place of the univariate scale-averaged power. The resulting continuous strength measure is used exactly as in the univariate case, as a multiplicative position-size scalar.

6.2 Regime Detection via Hidden Markov Models

The distinction between "a cycle is statistically present" and "the market is calm or turbulent" is not obvious *a priori*, and the wavelet machinery of Section 2.6 is explicitly a test of the former, not the latter. To test the relationship between the two formally, a two-state Gaussian Hidden Markov Model [6, 7] is fit to the commodity's log-returns, with the higher-variance state labeled "stressed" and the lower-variance state labeled "calm". Because expectation-maximization is invariant to an arbitrary relabeling of the latent states, the higher-variance state is identified *after* each fit by its emission variance, never by a fixed state index.

Exactly as for the wavelet signal, a fully causal walk-forward version is built: at each refit point, a fresh HMM is estimated using only the expanding window of data available up to that point, and the smoothed posterior probability of the "stressed" state *at the window's last observation* is retained as the live regime signal. As in Section 4.3, this endpoint evaluation is causal because the window itself never extends past "today", even though the forward-backward smoother technically uses the whole window internally.

This regime probability is used in two structurally different ways, described next.

6.3 Regime Gating (Directional Use)

The most direct hypothesis – trade the wavelet mean-reversion signal *more aggressively* when the HMM detects a stressed regime, on the premise that this is when the wavelet-detected cyclicality is strongest – is tested by multiplying position size by the HMM stress probability, in the same way as the wavelet strength multiplier. As reported in Section 7, this directional use of the regime signal **underperforms** not using it at all.

6.4 Risk-Reduction Overlay (Non-Directional Use)

The alternative hypothesis is that the same regime information is useful, but only when used to *reduce*, not increase, exposure during detected stress, regardless of the direction of any underlying bet:

$$\tilde{r}_n^{\text{overlay}} = \tilde{r}_n \times \big(1 - \kappa \cdot \text{clip}(\hat{p}_n^{\text{stress}}, 0, 1)\big), \tag{17}$$

where $\hat{p}_n^{\text{stress}}$ is the causal HMM stress probability and $\kappa \in [0, 1]$ controls the strength of the de-risking (this study uses $\kappa = 1$, i.e. exposure can be reduced to zero at maximum detected stress). This is a genuinely different use of the same underlying information: it makes no bet on *what* will happen during stress, only on the (much weaker, and more defensible) claim that being smaller during turbulence is prudent. Equation (17) is applied, separately, to two very different base exposures: a naive long-only buy-and-hold position, and the diversified band ensemble of

Section 5, in order to test whether the overlay's effect depends on the nature of the underlying exposure.

7 Empirical Results

All results below are for WTI crude oil futures (`CL=F`), 2 January 2014 to 4 September 2026 (3188 observations), using a primary band of 20–60 trading days.

7.1 Descriptive Wavelet and Coherence Analysis

Before turning to strategy performance, Figure 1 shows the full-sample (non-causal, purely descriptive – see Section 3.2) wavelet power spectrum of CL=F log-returns. The most conspicuous feature is a broad, high-power region centered around the 128–256 day scale in the middle of the sample, coinciding with the COVID-19 demand shock and the negative-price event of April 2020; a second, weaker elevated region appears near the right edge of the sample. Shorter-period power (below roughly 16 days) is present throughout but fragmented and rarely significant, consistent with the low average wavelet strength (0.21 on the 0–1 scale of Section 4) reported later.

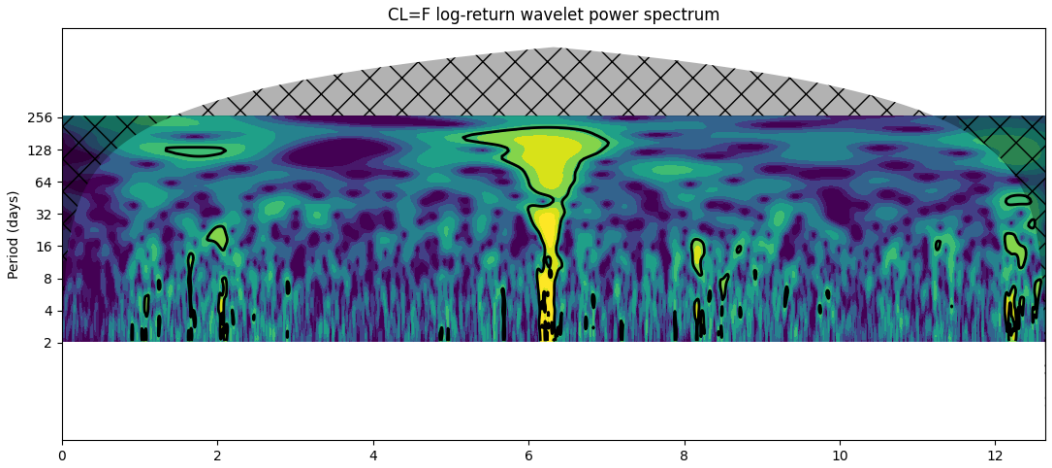
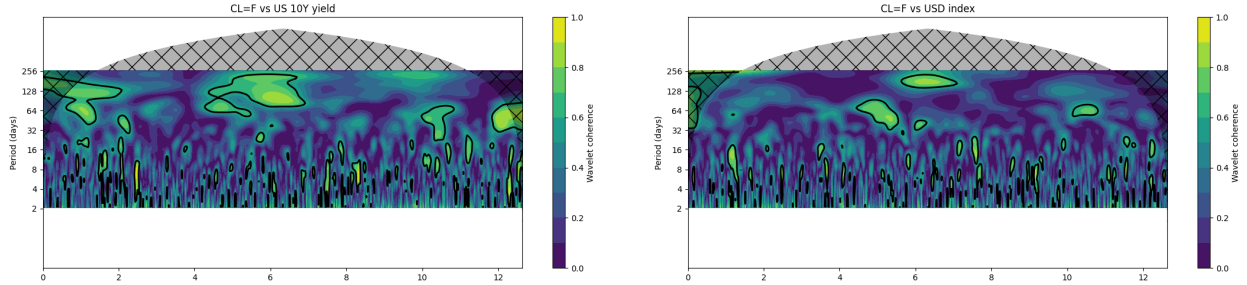


Figure 1: Full-sample wavelet power spectrum of CL=F log-returns (Morlet, $\omega_0 = 6$). Black contours enclose regions significant at the 95% level against a red-noise background (Section 2.6); the cross-hatched region is the cone of influence (Section 2.5), where edge effects make the estimate unreliable. The dominant feature, centered near periods of 128–256 days around year 6 of the sample (2020), coincides with the COVID-19 shock and the negative-price event of 20 April 2020.

Figure 2 shows the full-sample wavelet coherence (Section 2.8, with Monte Carlo significance testing following Grinsted, Moore, and Jevrejeva [2]) between CL=F and two macroeconomic series: the US 10-year Treasury yield and the US dollar index. Both show patches of significant coherence concentrated at longer periods (64–256 days), consistent with a slow-moving, macro-driven relationship rather than a high-frequency one; neither shows the kind of persistent, sample-spanning coherence that would justify using either series as a reliable short-term leading indicator, a conclusion corroborated by the negative walk-forward results for both drivers in Section 6.1 and Table 4.



(a) CL=F vs. US 10-year Treasury yield.

(b) CL=F vs. USD index.

Figure 2: Full-sample wavelet coherence between CL=F log-returns and two macroeconomic drivers. Colour indicates coherence (0–1); black contours enclose regions significant at the 95% level via Monte Carlo surrogate testing (Section 2.8); the cross-hatched region is the cone of influence.

7.2 Wavelet Strategy and Buy-and-Hold Benchmark

Table 1: Primary band strategy vs. buy-and-hold benchmark, full sample.

	Sharpe	Ann. Return	Ann. Vol.	Max Drawdown
Wavelet strategy (20–60d)	−0.02	−0.1%	4.7%	19.9%
Buy & hold	0.24	11.2%	47.1%	90.7%

The single-band strategy shows no distinguishable edge in isolation, but achieves this at roughly one-tenth the buy-and-hold's volatility and less than a quarter of its drawdown – already a first hint that the value of this machinery may lie in risk shaping rather than directional prediction.

7.3 Out-of-Sample Band Selection and the Instability Problem

Selecting the best band on a single training period (through May 2021) and testing it on the untouched remainder produces the pattern in Table 2.

Table 2: Out-of-sample band selection, single split (train ends 2021-05-28).

Band (days)	Train Sharpe	Test Sharpe	Test Ann. Ret.	Test Max DD
5–20	0.18	0.74	3.7%	5.8%
10–30	−0.07	0.85	4.8%	4.8%
20–60	−0.43	0.49	2.5%	8.8%
60–120	0.40	0.13	0.6%	10.8%

The band chosen *ex ante* on the training data (60–120 days) is, in fact, the *worst*-performing band on the test data; the bootstrap 90% confidence interval for its test Sharpe, [−0.42, 0.73], straddles zero. Repeating this exercise across nine rolling annual windows (2016–2024) confirms this is not an artifact of one unlucky date: the ex-ante pick ranked in the top half of the actual test results in only 4 of 9 windows – statistically indistinguishable from chance – although it did achieve a positive test Sharpe in 6 of 9 windows.

7.4 Ensemble Diversification

The equal-weight ensemble across all four candidate bands, which requires no ex-ante selection at all, outperformed the ex-ante-selected single band in 6 of the same 9 rolling windows (average test Sharpe 0.38 vs. 0.12), and its full-sample statistics are reported in Table 3.

Table 3: Equal-weight ensemble across four bands, full sample.

	Sharpe	Ann. Return	Ann. Vol.	Max Drawdown
Ensemble	0.43	1.4%	3.3%	8.4%

Block bootstrap 90% CI: [0.02, 0.83], Pr(Sharpe > 0) = 96% – the first result in this study whose confidence interval excludes zero.

7.5 Cross-Wavelet Macro Drivers

Gating the primary band's signal by cross-wavelet strength against four macroeconomic drivers (Table 4) shows **no** consistent full-sample improvement over either an ungated signal or the univariate significance gate; where individual drivers show a better *test*-period Sharpe, it is concentrated in the same 2020–2024 window examined in Section 8, and is not corroborated by the full-sample statistic.

Table 4: Strength-source comparison, primary band, full sample and single train/test split.

Variant	Sharpe	Ann. Ret.	Max DD	Train Sharpe	Test Sharpe
No gate	0.13	1.7%	55.9%	−0.61	0.99
Self-significance gate	−0.02	−0.1%	19.9%	−0.43	0.49
USD index (xwt)	−0.15	−0.6%	25.6%	−0.66	0.92
US 10Y yield (xwt)	−0.34	−1.9%	34.5%	−0.63	0.91
VIX (xwt)	−0.10	−0.5%	24.9%	−0.53	1.04
S&P 500 (xwt)	−0.03	−0.2%	25.7%	−0.46	1.13

7.6 HMM Regime Detection

Fitting the two-state HMM of Section 6.2 over the full sample and correlating its smoothed stress probability against the wavelet strength signal of the primary band gives a correlation of **0.61** (evaluated post warm-up): two independently constructed statistical tools – a spectral significance test and a latent-state regime classifier – agree that the same historical periods are "special". Using the HMM signal **directionally**, however (as an additional gate on the same mean-reversion trade), is counterproductive (Table 5).

Table 5: Directional use of the HMM regime signal, primary band.

Variant	Sharpe	Ann. Ret.	Max DD	Train Sharpe	Test Sharpe
No gate	0.13	1.7%	55.9%	−0.61	0.99
Wavelet self-sig. gate	−0.02	−0.1%	19.9%	−0.43	0.49
HMM regime gate only	−0.26	−1.8%	36.8%	−0.53	0.32
HMM × wavelet gate	−0.17	−0.6%	16.6%	−0.44	0.26

7.7 Risk-Reduction Overlay

Applying the *non-directional* overlay of Eq. (17) to two different base exposures produces sharply divergent results (Table 6).

Table 6: Risk-reduction overlay ($\kappa = 1$), full sample.

Base	Overlay	Sharpe	Ann. Ret.	Max DD	Calmar
Buy & hold	none	0.24	11.2%	90.7%	0.12
Buy & hold	wavelet strength	0.03	0.9%	72.8%	0.01
Buy & hold	HMM stress	0.08	2.3%	75.9%	0.03
Ensemble	none	0.43	1.4%	8.4%	0.17
Ensemble	wavelet strength	0.70	1.2%	5.8%	0.21
Ensemble	HMM stress	0.61	1.1%	4.9%	0.23

On the ensemble, both overlays *improve* Sharpe, Calmar, and maximum drawdown simultaneously, and the rolling window check confirms the effect is broad rather than concentrated in a single period: maximum drawdown was reduced in 8 of 9 rolling windows for the HMM-stress overlay on the ensemble (7 of 9 on buy-and-hold). On buy-and-hold, the same overlay reduces drawdown but at the cost of most of the average return, so both Sharpe and Calmar *worsen*. Section 8 explains why.

Figure 3 brings the full arc of the empirical section together on a single time axis, from the raw price series down to the risk-overlay equity curve.

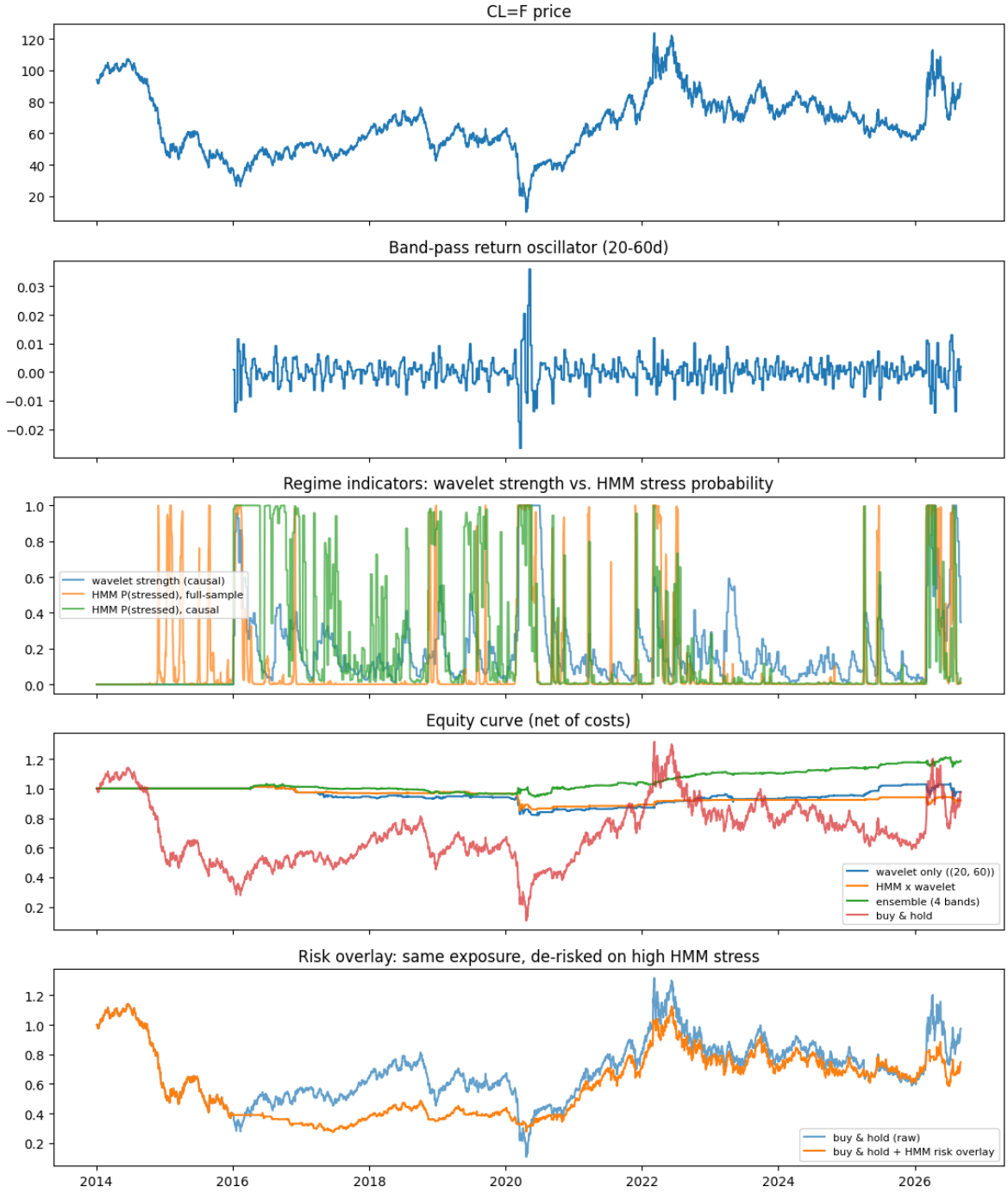


Figure 3: Summary of the empirical pipeline on CL=F, 2014–2026, all panels sharing the same time axis. **(a)** Raw price. **(b)** Causal walk-forward band-pass return oscillator, 20–60 day band (Section 4.3); note the pronounced amplitude spike during the 2020 shock. **(c)** The two independently constructed regime indicators overlaid: causal wavelet strength (blue), full-sample HMM stress probability (orange, descriptive only), and causal walk-forward HMM stress probability (green) – their broad co-movement is the basis for the 0.61 correlation reported in Section 6.2. **(d)** Net-of-cost equity curves for the single-band strategy, the HMM×wavelet directional gate, the four-band ensemble, and buy-and-hold: the ensemble is the only variant combining a smooth equity curve with a positive long-run slope. **(e)** The risk-reduction overlay of Eq. (17) applied to buy-and-hold: drawdown is visibly reduced around 2015–2016 and 2020, but at a proportional cost to the recoveries that follow, illustrating the asymmetry discussed in Section 8.

8 Discussion

8.1 Cyclicality Is Not Volatility

The single most important empirical finding of this study is the distinction, confirmed by two independent tools, between *"the market is turbulent"* and *"the market is cyclical (mean-reverting)"*. These two notions are correlated (0.61 in Section 6.2) but not interchangeable, and the difference has direct trading consequences. The COVID-19 collapse and the 2022 Russia/Ukraine supply shock are periods of extreme variance, but they are episodes of strongly *directional*, one-off price adjustment – not repeating oscillations. A wavelet transform, being a decomposition of *energy* across frequencies, can register a sharp V-shaped move (a collapse followed by a partial recovery) as substantial power across a wide range of scales, even though the event is a single, non-repeating structural shift rather than a genuine cycle. Using this elevated "strength" reading to increase a mean-reversion bet is precisely the wrong response to a trending market, which explains why every directional regime-gating variant in Tables 5 underperformed the ungated baseline: it consistently sized up mean-reversion exposure exactly when the market was least likely to revert.

8.2 Why the Overlay Works on One Exposure but Not the Other

The asymmetric result in Table 6 has a clear economic explanation rather than being a statistical curiosity. A buy-and-hold position carries **persistent directional exposure**: de-risking during detected stress necessarily also de-risks during the sharp recoveries that, in commodity markets, frequently occur *while* realized volatility is still elevated (e.g. the rebound following the 2020 collapse). The overlay therefore systematically forfeits a disproportionate share of the position's total return, worsening risk-adjusted performance even as it reduces drawdown in isolation. The ensemble, by contrast, is a book of frequently-flipping mean-reversion bets with no persistent directional exposure to give up: reducing its size during turbulent periods avoids bad days without a corresponding opportunity cost from a rally the strategy was never structurally positioned to capture in full. This is a specific, empirically demonstrated instance of a broader and well-known caveat about volatility-targeting overlays: their value depends critically on whether the underlying exposure has a directional risk premium that de-risking would forfeit.

8.3 Multiple Testing and the Limits of This Study

Every positive result reported in Section 7 was obtained only after testing several bands, several macroeconomic drivers, and several regime-gating variants; the out-of-sample and rolling-window machinery of Section 5 was built specifically to guard against over-interpreting any single number that happens to look good. Even so, two limitations should be stated plainly. First, the strongest positive results (the ensemble and its overlay) are still concentrated, to a meaningful degree, in the 2020–2024 sub-period; while the rolling validation shows the drawdown-reduction effect to be broad, a longer sample spanning additional volatility regimes (or the same pipeline applied to other commodities, as briefly explored during this study for corn and natural gas without a stable cross-commodity pattern emerging) would be needed to state the finding with full confidence. Second, the risk overlay of Eq. (17) scales an already-computed, cost-net return series rather than reworking the position-sizing step before transaction costs are computed; a fully rigorous implementation would fold the de-risking directly into position sizing, alongside the existing volatility target.

9 Conclusion

This study set out to test whether wavelet-based spectral decomposition can generate a stable, exploitable trading edge in commodity futures, with the statistical rigor – causal walk-forward estimation, closed-form and Monte Carlo significance testing, bootstrap confidence intervals, out-of-sample band selection, rolling multi-window validation – that the question deserves. The direct

answer is negative: no single band, driver, or regime-gating variant produced a directional signal with a confidence interval reliably excluding zero across the full sample and multiple independent validation windows.

The more useful answer, however, emerged from taking the negative results seriously rather than discarding them: wavelet-detected spectral power is formally confirmed (via an independently constructed Hidden Markov regime classifier) to concentrate in high-volatility periods, and this same information – used not as a directional bet but as a **risk-reduction overlay** – materially improves the risk-adjusted profile of a diversified, non-directional exposure while backfiring on a purely directional one, for a reason that follows directly from the economics of the two exposures. This distinction between *prediction* and *risk management*, and between *volatility* and *cyclicality*, is the substantive contribution of this report, and illustrates a general lesson about applying signal-processing tools developed for geophysics to financial markets: the tool transplants cleanly, but the null hypothesis it should be tested against, and the use to which a "significant" reading should be put, do not.

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